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pre-miRBench: a multispecies benchmark and reference model for animal precursor microRNA prediction

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1 INTRODUCTION

71 animal species (MirGeneDB 3.0)

77,616 records: 7,056 pos / 70,560 neg

4 species × family test settings

7 predictors benchmarked

Published pre-miRNA predictors use different species, negatives and data splits, making direct comparison difficult. Prior benchmarks did not cross species holdouts with named miRNA-family holdouts; pre-miRBench combines that split design with curated MirGeneDB annotations and classifier-mined hard negatives, providing a complete benchmark dataset and a reference model trained on it.

2 METHODS

- **Positives:** 71 MirGeneDB 3.0 species with a matching genome assembly; 200-nt windows centered on curated precursor loci. Windows with >10% soft-masked or N bases were excluded (7,056 final positives).
- **Candidate negatives:** genome scanned with 200-nt windows at a 50-nt stride; Candidates folded with RNAfold and kept only if they pass four permissive hairpin filters: MFE ≤ −10 kcal/mol, paired fraction ≥ 0.40, contiguous stem ≥ 8 bp, unpaired run ≤ 25 nt.
- **Hard-negative mining:** per species, a 10-member random-forest ensemble trained using only that species' training positives; kept candidates with P(positive) ≥ 0.5. Fixed 1:10 positive:negative ratio throughout.
- **Duplicate removal:** every record normalized and center-cropped to the shortest model input (100 nt) to identify repeated inputs.

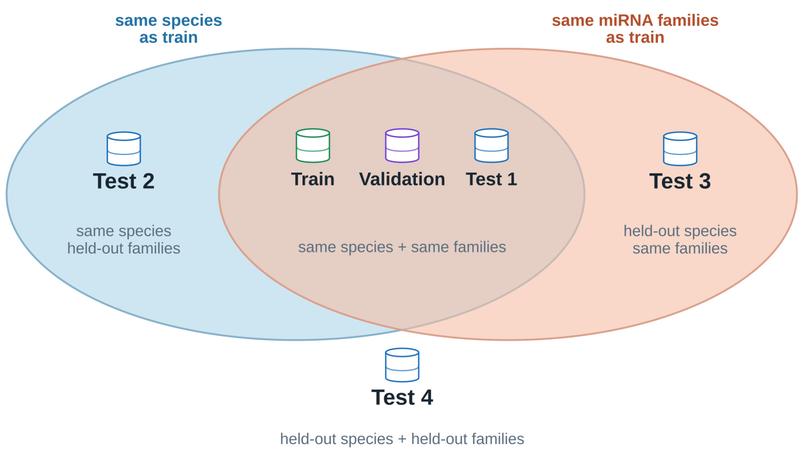


Fig. 1 — pre-miRBench split design. Training/validation share species and families; the four test sets cross species status with family status.

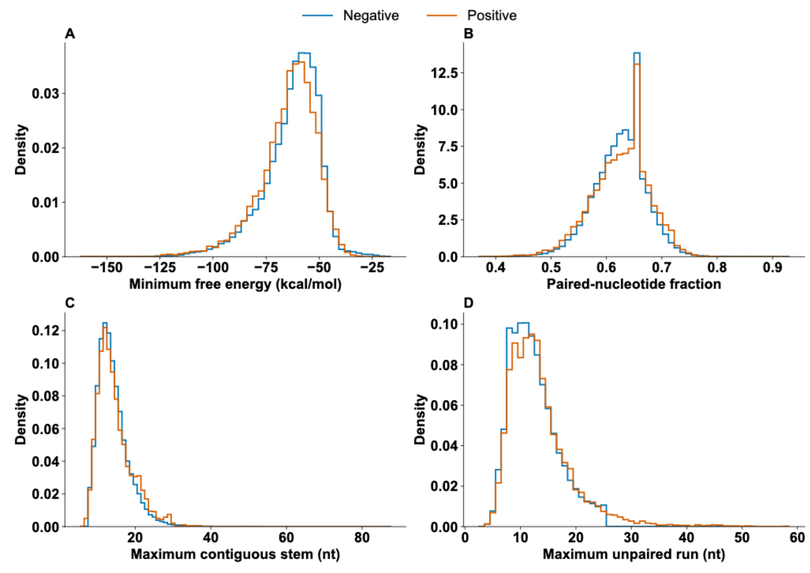


Fig. 2 — RNAfold feature distributions in the final benchmark

MODELS COMPARED

Six published predictors, retrained under one protocol

DeepMir sequence CNN	**deepMiRGene** hybrid deep net	**dnnPreMiR** sequence+structure CNN
mirDNN residual CNN	**miRe2e** transformer	**MuStARD** multi-branch CNN

The pre-miRBench model ensemble of one PairMessageCNN and two SpeciesGraphGRU checkpoints (1,178,309 parameters); developed autonomously with **Agentomics** 1.0.4 (gpt-5.6-sol) — an LLM-agent system for biomedical ML — using validation AP for model selection. The components cover local sequence and structure patterns, interactions between paired positions, recurrent context across the window and global folding summaries.

EXPERIMENTAL SETUP

- **Primary metric: Average Precision (AP)** over the full ranked list. Appropriate for the fixed 1:10 class ratio; reflects the trade-off between recovering positives and limiting false-positive predictions.
- **Secondary metrics**: AUROC, accuracy / precision / recall / F1 / MCC at threshold 0.5.

3 RESULTS

Average Precision (AP) across generalization settings — Table 1

Predictor	Known/known	Known/held-out	Held-out/known	Held-out/held-out
DeepMir	0.8648	0.8513	0.8685	0.8020
deepMiRGene	0.9648	0.9703	0.9729	0.9422
dnnPreMiR	0.9782	0.9595	0.9821	**0.9677**
mirDNN	0.9845	0.9561	0.9853	0.9113
miRe2e	0.9628	0.9499	0.9532	0.8685
MuStARD	0.9815	0.9696	0.9789	0.9526
pre-miRBench	**0.9869**	**0.9813**	**0.9895**	0.9659

Values are AP; highest per column is bold. The convention used is species/family.

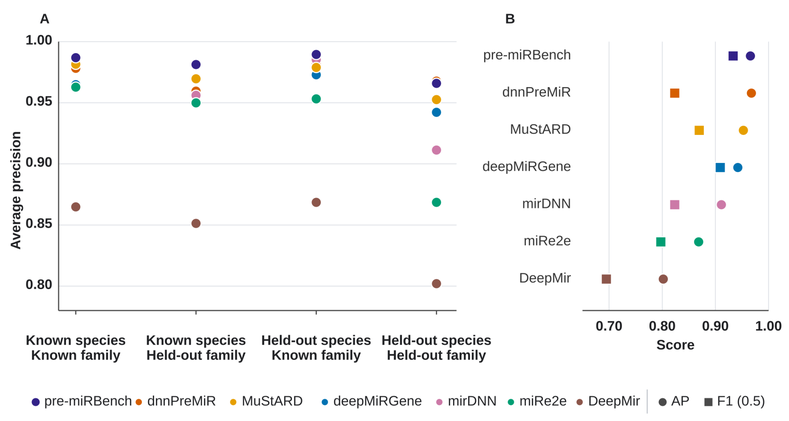


Fig. 3 — (A) AP across the four test settings. (B) AP and F1 (@0.5) on the held-out-species/held-out-family test.

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→ GENERALIZATION RESULTS

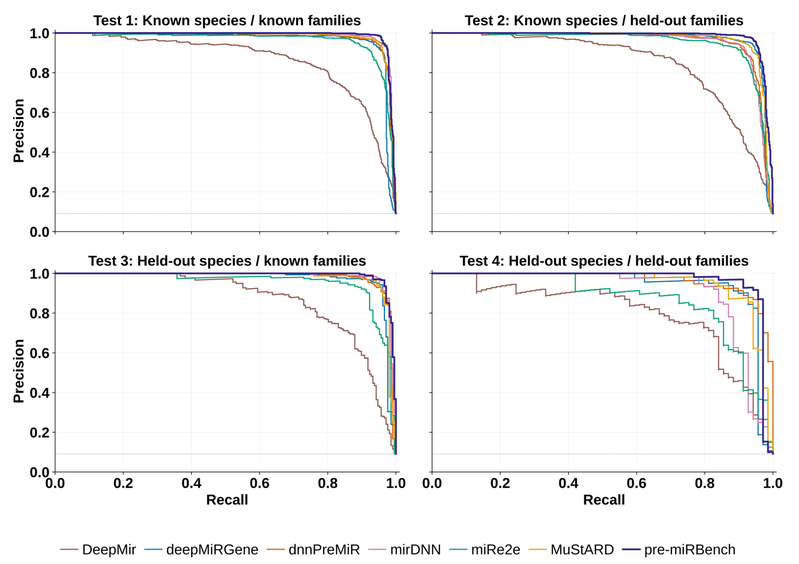


Fig. 4 — precision–recall, all seven predictors, four test settings.

KEY FINDING

pre-miRBench had the highest mean AP across the four tests (**0.9809**), ranking first in three; **dnnPreMiR** led the joint species-and-family holdout (0.9677 vs. 0.9659; ΔAP = −0.0018, 95% CI −0.0445 to 0.0341).

4 DISCUSSION & CONCLUSION

Species and family holdouts remove different information: a species holdout can still contain families seen elsewhere, while a family holdout can contain species that contributed other precursors. AP did not fall consistently when either factor was held out alone.

Limitations

- The joint holdout covers only two species (Gallus gallus and Drosophila melanogaster; 69 positives / 690 negatives), so its confidence interval is wide relative to the AP gap between the top two models.
- Benchmark prevalence is 1:10; genome scanning is substantially more imbalanced and may yield more false-positive calls.
- Negatives are annotation-derived genomic controls, not experimentally confirmed non-miRNAs; whole-genome false-discovery remains unknown.
- More held-out species and similarity-clustered families are needed to resolve model differences and measure transfer across broader evolutionary distances.

pre-miRBench provides a 71-species dataset, four fixed species-and-family test settings, reproducible implementations for six published predictors and a new reference model. The fixed test sets, trained weights and record-level predictions provide a reproducible reference for evaluating future precursor-miRNA predictors.

AVAILABILITY & CONTACT

Code: **github.com/dimostzim/pre-miRBench**

Data, weights & predictions: **zenodo.org/records/22813044**

Scan for the preprint, poster PDF, code and data.

Contact: vchatzit@di.uoa.gr

Selected references

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